

Medicine Outcome Prediction System

SHAP model explainability | Held-out test-set interpretation

Purpose

SHAP values explain how each clinical feature moves the predicted probability of a Yes treatment outcome. Models are compared using the same report split; explanations use a reproducible sample of the held-out test set.

Important interpretation

A large SHAP value indicates influence on model output, not causation or treatment effectiveness. Feature importance can reflect correlations, missingness patterns, and the training data distribution. Use these results for audit and hypothesis generation, not independent clinical decisions.

Analysis settings

Models: 4

Background rows: 25

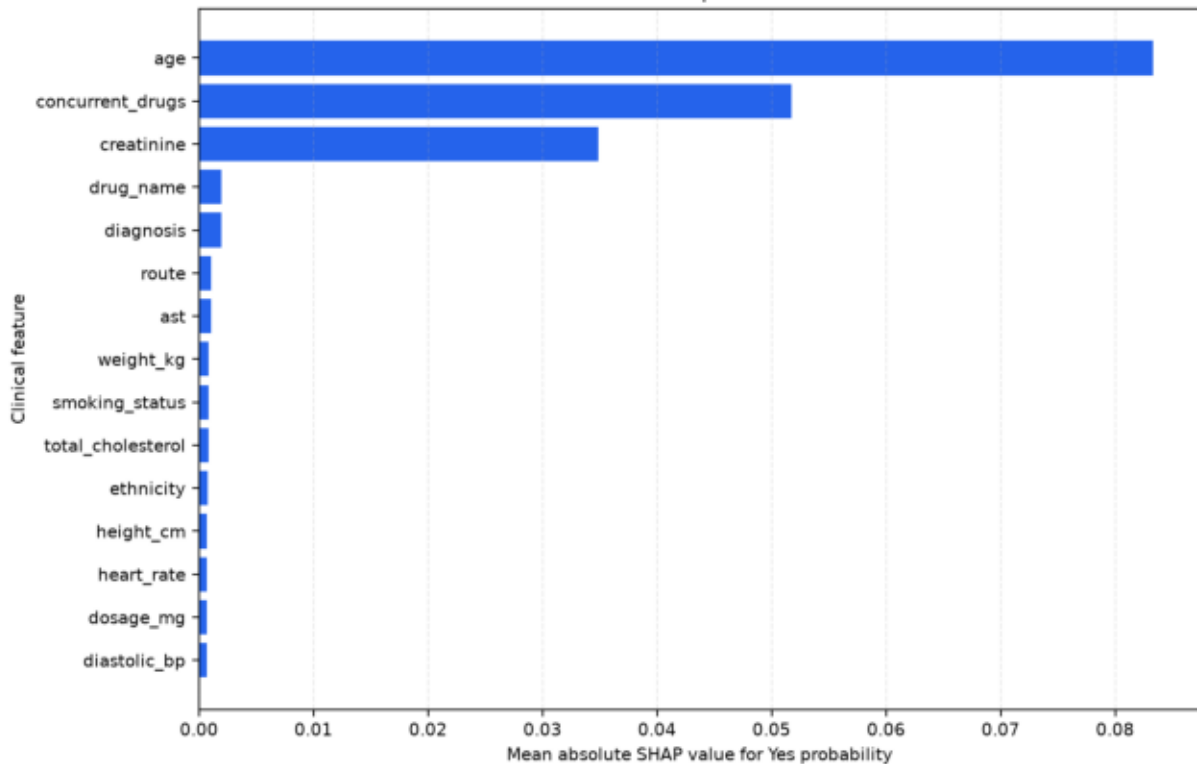
Explained test rows per model: 50

Target explained: positive-class probability (treatment_outcome = Yes)

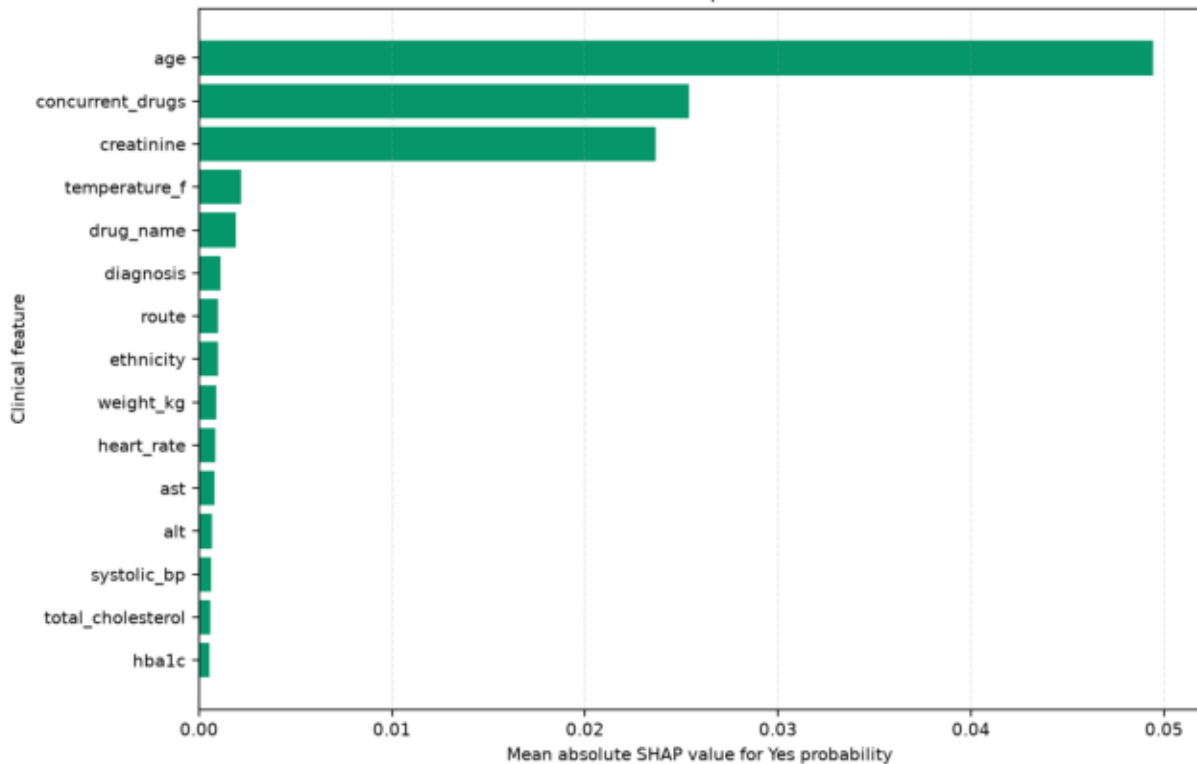
Feature names: one-hot encoded columns aggregated back to original clinical variables

The detailed feature rankings and model-specific plots are saved in the Outputs folder.

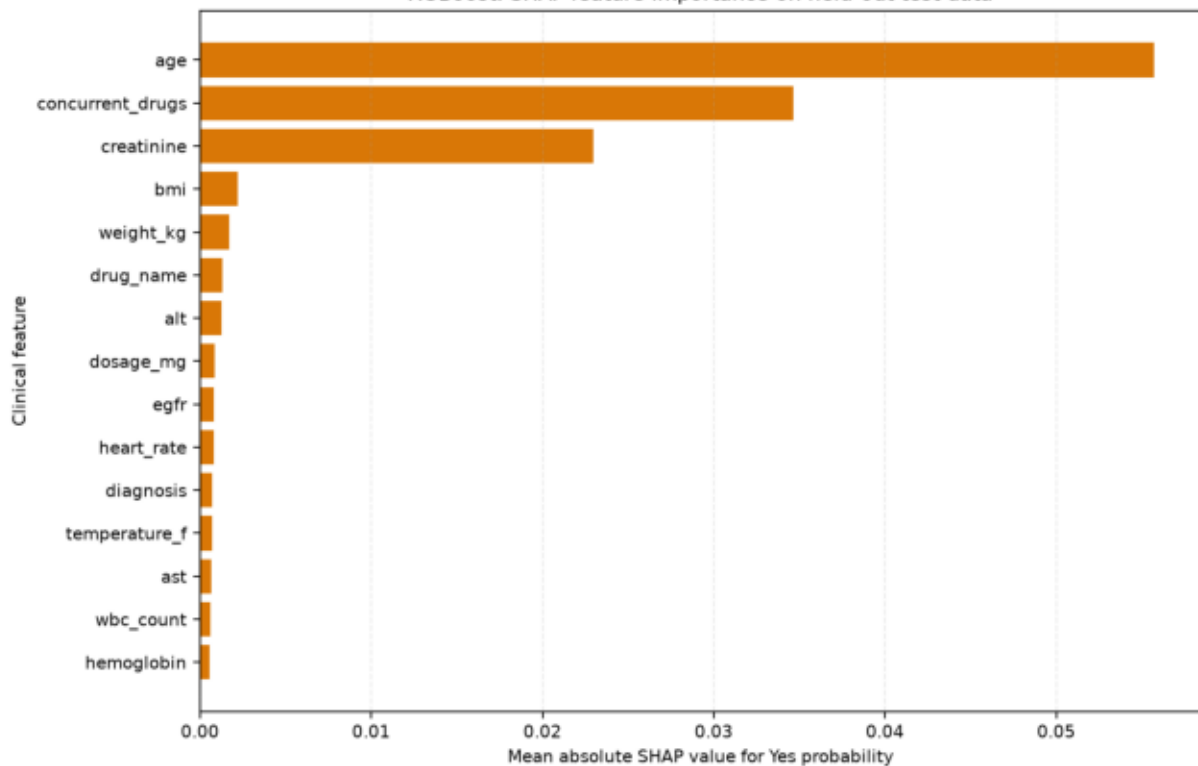
Decision Tree: SHAP feature importance on held-out test data



Random Forest: SHAP feature importance on held-out test data



XGBoost: SHAP feature importance on held-out test data



Gradient Boosting: SHAP feature importance on held-out test data

